

2.3 Change of Parameter; Arc Length Worksheet

1. Determine if the following functions are smooth.

a) $\mathbf{r}(t) = \left\langle \frac{1}{2}t^2 - t, \frac{1}{3}t^3 + \frac{1}{2}t^2 - 2t, \ln(t) \right\rangle$ where $0 < t < \infty$.

We see that $\mathbf{r}'(t) = \left\langle t-1, t^2+t-2, \frac{1}{t} \right\rangle$

Clearly each component is continuous on $(0, \infty)$ and since $\frac{1}{t} \neq 0$ for any t then $\mathbf{r}'(t) \neq \mathbf{0}$ for any t . So, $\mathbf{r}(t)$ is smooth.

b) $\mathbf{r}(t) = \cos(t+1)\mathbf{i} + (e^{t+1} - t)\mathbf{j} + \mathbf{k}$

We see that $\mathbf{r}'(t) = -\sin(t+1)\mathbf{i} + (e^{t+1} - 1)\mathbf{j} + 0\mathbf{k}$

Clearly each component is continuous on $(-\infty, \infty)$, but since when $t = -1$, it is the case that $\mathbf{r}'(t) = \mathbf{0}$ we conclude that $\mathbf{r}(t)$ is not smooth.

c) $\mathbf{r}(t) = \sin(t)\mathbf{i} + \cos(t)\mathbf{j}$

We see that $\mathbf{r}'(t) = \cos(t)\mathbf{i} - \sin(t)\mathbf{j}$

Clearly each component is continuous on $(-\infty, \infty)$ and since sine and cosine share not zeros for any t then $\mathbf{r}'(t) \neq \mathbf{0}$ for any t . So, $\mathbf{r}(t)$ is smooth.

d) $\mathbf{r}(t) = \langle \sqrt{t}, t, t \rangle$ where $0 \leq t \leq 2$.

We see that $\mathbf{r}'(t) = \left\langle \frac{1}{2\sqrt{t}}, 1, 1 \right\rangle$

Clearly it is impossible for $\mathbf{r}'(t) = \mathbf{0}$ for any t , however $\frac{1}{2\sqrt{t}}$ is discontinuous at $t = 0$. Therefore, we conclude that $\mathbf{r}(t)$ is not smooth.

2. Find the length of the curve given by $\mathbf{r}(t) = \langle 2, 2t^2, t^3 \rangle$ for $1 \leq t \leq 2$.

We see that $\mathbf{r}'(t) = \langle 0, 4t, 3t^2 \rangle$ and thus $\|\mathbf{r}'(t)\| = \sqrt{16t^2 + 9t^4}$. Clearly we see that $\mathbf{r}'(t)$ is continuous on $1 \leq t \leq 2$ and $\mathbf{r}'(t) \neq \mathbf{0}$ for any $1 \leq t \leq 2$.

Therefore, we know that the length of the curve is given by:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_1^2 \sqrt{16t^2 + 9t^4} dt = \int_1^2 t\sqrt{16 + 9t^2} dt$$

$$\text{Let } u = 16 + 9t^2 \text{ which implies } du = 18tdt \quad \frac{1}{18} du = tdt.$$

$$\text{Our bounds become: } u(1) = 25 \text{ \& } u(2) = 52.$$

$$\text{So, } L = \int_1^2 t\sqrt{16 + 9t^2} dt = \int_{25}^{52} \frac{1}{18} \sqrt{u} du = \frac{1}{27} u^{3/2} \Big|_{25}^{52} = \frac{1}{27} [52^{3/2} - 25^{3/2}] = \frac{1}{27} [52^{3/2} - 125]$$

3. Consider the curve $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ for $0 \leq t \leq 6\pi$.

- a) Find the length of $\mathbf{r}(t)$.

We see that $\mathbf{r}'(t) = \langle -\sin(t), \cos(t), 1 \rangle$ and thus $\|\mathbf{r}'(t)\| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$. Clearly we see that $\mathbf{r}'(t)$ is continuous on $0 \leq t \leq 6\pi$ and $\mathbf{r}'(t) \neq \mathbf{0}$ for any $0 \leq t \leq 6\pi$.

Therefore, we know that the length of the curve is given by:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_0^{6\pi} \sqrt{2} dt = \sqrt{2} t \Big|_0^{6\pi} = 6\pi\sqrt{2}$$

- b) Create an arc length parameterization for the curve starting from when $t = 0$.

$$\text{We first determine the arc length from 0 to } t \text{ which is given by } s(t) = \int_0^t \sqrt{2} du = \sqrt{2}t.$$

$$\text{We then see that if } s = \sqrt{2}t \text{ then } t = \frac{s}{\sqrt{2}}.$$

Therefore, the arc length parameterization for the curve is:

$$\mathbf{r}(s) = \left\langle \cos\left(\frac{s}{\sqrt{2}}\right), \sin\left(\frac{s}{\sqrt{2}}\right), \frac{s}{\sqrt{2}} \right\rangle$$

- c) What point do you reach if travel along the curve $\mathbf{r}(t)$ for $\pi\sqrt{2}$ units, starting from when $t = 0$?

Here we are simply evaluating $\mathbf{r}(s)$ when $s = \pi\sqrt{2}$. We see that:

$$\mathbf{r}(\pi\sqrt{2}) = \langle \cos(\pi), \sin(\pi), \pi \rangle = \langle -1, 0, \pi \rangle.$$

Therefore, after moving $\pi\sqrt{2}$ units along $\mathbf{r}(t)$ we will have arrived at the point $(-1, 0, \pi)$.

4. Consider the curve $\mathbf{r}(t) = \left\langle t^2, \frac{8}{3}t^{3/2}, 4t \right\rangle$ for $0 \leq t \leq 10$.

a) Find the length of $\mathbf{r}(t)$.

We see that $\mathbf{r}'(t) = \langle 2t, 4\sqrt{t}, 4 \rangle$ and thus $\|\mathbf{r}'(t)\| = \sqrt{4t^2 + 16t + 16}$. Clearly we see that $\mathbf{r}'(t)$ is continuous on $t \geq 0$ and $\mathbf{r}'(t) \neq \mathbf{0}$ for any $t \geq 0$.

Therefore, we know that the length of the curve is given by:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_0^{10} \sqrt{4t^2 + 16t + 16} dt = \int_0^{10} \sqrt{4(t^2 + 4t + 4)} dt = \int_0^{10} \sqrt{4(t+2)^2} dt = \int_0^{10} 2|t+2| dt$$

Since we know that for $0 \leq t \leq 10$ the value of $t+2$ will always be positive, we can drop the absolute value bars.

$$\int_0^{10} 2|t+2| dt = \int_0^{10} (2t+4) dt = (t^2 + 4t) \Big|_0^{10} = (100 + 40) = 140$$

b) Create an arc length function for the curve starting from when $t = 5$.

We first determine the arc length from 5 to t which is given by

$$s(t) = \int_5^t \sqrt{4u^2 + 16u + 16} du = \dots = \int_5^t (2u+4) du = (u^2 + 4u) \Big|_5^t = (t^2 + 4t) - (25 + 20) = t^2 + 4t - 45$$

We then see that $s(t) = t^2 + 4t - 45$.

c) Explain why it would be difficult to create an arc length parameterization for this curve starting from when $t = 5$.

As we see above, $s(t) = t^2 + 4t - 45$. In order to create an arc length parameterization for $\mathbf{r}(t)$ we would need to solve the given function for t and then substitute the resulting expression in for t with the $\mathbf{r}(t)$ vector function. However, since solving $s(t) = t^2 + 4t - 45$ would require the use of the quadratic formula and would produce two different possible solutions for t , it is difficult to determine an arc length parameterization for $\mathbf{r}(t)$.

Note: We did not say that it is impossible to create an arc length parameterization for $\mathbf{r}(t)$, only that it would be more difficult than previous examples.

5. Consider the curve $\mathbf{r}(t) = \langle e^{2t}, e^{-2t}, 2\sqrt{2}t \rangle$ for $0 \leq t \leq 5$.

a) Find the length of $\mathbf{r}(t)$.

We see that $\mathbf{r}'(t) = \langle 2e^{2t}, -2e^{-2t}, 2\sqrt{2} \rangle$ and thus $\|\mathbf{r}'(t)\| = \sqrt{4e^{4t} + 4e^{-4t} + 8}$. Clearly we see that $\mathbf{r}'(t)$ is continuous on $0 \leq t \leq 5$ and $\mathbf{r}'(t) \neq \mathbf{0}$ for any $0 \leq t \leq 5$.

Therefore, we know that the length of the curve is given by:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_0^5 \sqrt{4e^{4t} + 4e^{-4t} + 8} dt = \int_0^5 \sqrt{4(e^{4t} + 2 + e^{-4t})} dt = \int_0^5 \sqrt{4(e^{2t} + e^{-2t})^2} dt = \int_0^5 2|e^{2t} + e^{-2t}| dt$$

Since we know that for $0 \leq t \leq 5$ the value of $e^{4t} + e^{-4t}$ will always be positive, we can drop the absolute value bars.

$$\int_0^5 2|e^{2t} + e^{-2t}| dt = \int_0^5 2(e^{2t} + e^{-2t}) dt = 2 \left(\frac{1}{2} e^{2t} - \frac{1}{2} e^{-2t} \right) \Big|_0^5 = e^{10} - e^{-10}$$

b) Create an arc length function for the curve with starting from $t = 0$.

We first determine the arc length from 0 to t which is given by

$$\begin{aligned} s(t) &= \int_0^t \sqrt{4e^{4u} + 4e^{-4u} + 8} du = \dots = \int_0^t 2(e^{2u} + e^{-2u}) du = 2 \left(\frac{1}{2} e^{2u} - \frac{1}{2} e^{-2u} \right) \Big|_0^t \\ &= (e^{2t} - e^{-2t}) - (1 - 1) = e^{2t} - e^{-2t} \end{aligned}$$

We then see that $s(t) = e^{2t} - e^{-2t}$

c) Reparametrize $\mathbf{r}(t)$ so that it is traced in the opposite direction.

Currently we see that the curve $\mathbf{r}(t)$ is sketched as t varies from 0 to 5.

We want to create a new parameter, τ , such that as τ varies from 0 to 5, t will vary from 5 to 0.

Consider $t = 5 - \tau$. As τ varies from 0 to 5 clearly t will vary from 5 to 0.

Therefore, our reparameterization is: $\mathbf{r}(t(\tau)) = \langle e^{2(5-\tau)}, e^{-2(5-\tau)}, 2\sqrt{2}(5-\tau) \rangle$ where $0 \leq \tau \leq 5$

6. Consider the curve $\mathbf{r}(t) = \left\langle \frac{\sqrt{10}}{2}t^2, \frac{1}{3}t^3, 5t+14 \right\rangle$ for $0 \leq t \leq 2$.

a) Find the length of $\mathbf{r}(t)$.

We see that $\mathbf{r}'(t) = \langle \sqrt{10}t, t^2, 5 \rangle$ and thus $\|\mathbf{r}'(t)\| = \sqrt{10t^2 + t^4 + 25}$. Clearly we see that $\mathbf{r}'(t)$ is continuous on $0 \leq t \leq 2$ and $\mathbf{r}'(t) \neq \mathbf{0}$ for any $0 \leq t \leq 2$.

Therefore, we know that the length of the curve is given by:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_0^2 \sqrt{10t^2 + t^4 + 25} dt = \int_0^2 \sqrt{(t^2 + 5)^2} dt = \int_0^2 |t^2 + 5| dt$$

Since we know that the value of $t^2 + 5$ is always positive, we can drop the absolute value bars.

$$\int_0^2 |t^2 + 5| dt = \int_0^2 (t^2 + 5) dt = \left(\frac{1}{3}t^3 + 5t \right) \Big|_0^2 = \frac{8}{3} + 10 = \frac{38}{3}$$

b) Create an arc length function for the curve with starting from $t = 0$.

We first determine the arc length from 0 to t which is given by

$$s(t) = \int_0^t \sqrt{10u^2 + u^4 + 25} du = \dots = \int_0^t (u^2 + 5) du = \left(\frac{1}{3}u^3 + 5u \right) \Big|_0^t = \frac{1}{3}t^3 + 5t.$$

We then see that $s(t) = \frac{1}{3}t^3 + 5t$.

c) Reparametrize $\mathbf{r}(t)$ so that it is traced in the opposite direction for $10 \leq \tau \leq 20$

Currently we see that the curve $\mathbf{r}(t)$ is sketched as t varies from 0 to 2.

We want to create a new parameter, τ , such that as τ varies from 10 to 20, t will vary from 2 to 0.

Consider $t = 4 - \frac{1}{5}\tau$. As τ varies from 10 to 20 clearly t will vary from 2 to 0.

Therefore, our reparameterization is: $\mathbf{r}(t(\tau)) = \left\langle \frac{\sqrt{10}}{2} \left(4 - \frac{1}{5}\tau \right)^2, \frac{1}{3} \left(4 - \frac{1}{5}\tau \right)^3, 5 \left(4 - \frac{1}{5}\tau \right) + 14 \right\rangle$

where $10 \leq \tau \leq 20$.

7. Consider the curve $\mathbf{r}(t) = \left\langle 3t, 4t, \frac{2}{3}t^{3/2} \right\rangle$ for $0 \leq t \leq 1$.

a) Find the length of $\mathbf{r}(t)$.

We see that $\mathbf{r}'(t) = \langle 3, 4, \sqrt{t} \rangle$ and thus $\|\mathbf{r}'(t)\| = \sqrt{25+t}$. Clearly we see that $\mathbf{r}'(t)$ is continuous on $0 \leq t \leq 1$ and $\mathbf{r}'(t) \neq \mathbf{0}$ for any $0 \leq t \leq 1$.

Therefore, we know that the length of the curve is given by:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_0^1 \sqrt{25+t} dt = \frac{2}{3}(25+t)^{3/2} \Big|_0^1 = \frac{2}{3}(26)^{3/2} - \frac{2}{3}(25)^{3/2} = \frac{2}{3}(26)^{3/2} - \frac{250}{3}$$

b) Create an arc length parameterization for the curve starting from when $t = 0$.

We first determine the arc length from 0 to t which is given by

$$s(t) = \int_0^t \sqrt{25+u} du = \frac{2}{3}(25+u)^{3/2} \Big|_0^t = \frac{2}{3}(25+t)^{3/2} - \frac{250}{3}.$$

We then see that if $s = \frac{2}{3}(25+t)^{3/2} - \frac{250}{3}$ then $t = \left[\frac{3}{2} \left(s + \frac{250}{3} \right) \right]^{2/3} - 25$.

Therefore, the arc length parameterization for the curve is:

$$\mathbf{r}(s) = \left\langle 3 \left(\left[\frac{3}{2} \left(s + \frac{250}{3} \right) \right]^{2/3} - 25 \right), 4 \left(\left[\frac{3}{2} \left(s + \frac{250}{3} \right) \right]^{2/3} - 25 \right), \frac{2}{3} \left(\left[\frac{3}{2} \left(s + \frac{250}{3} \right) \right]^{2/3} - 25 \right)^{3/2} \right\rangle$$

c) Reparametrize $\mathbf{r}(t)$ so that it is traced in the opposite direction.

Currently we see that the curve $\mathbf{r}(t)$ is sketched as t varies from 0 to 1.

We want to create a new parameter, τ , such that as τ varies from 0 to 1, t will vary from 1 to 0.

Consider $t = 1 - \tau$. As τ varies from 0 to 1 clearly t will vary from 1 to 0.

Therefore, our reparameterization is: $\mathbf{r}(t(\tau)) = \left\langle 3(1-\tau), 4(1-\tau), \frac{2}{3}(1-\tau)^{3/2} \right\rangle$ where $0 \leq \tau \leq 1$.